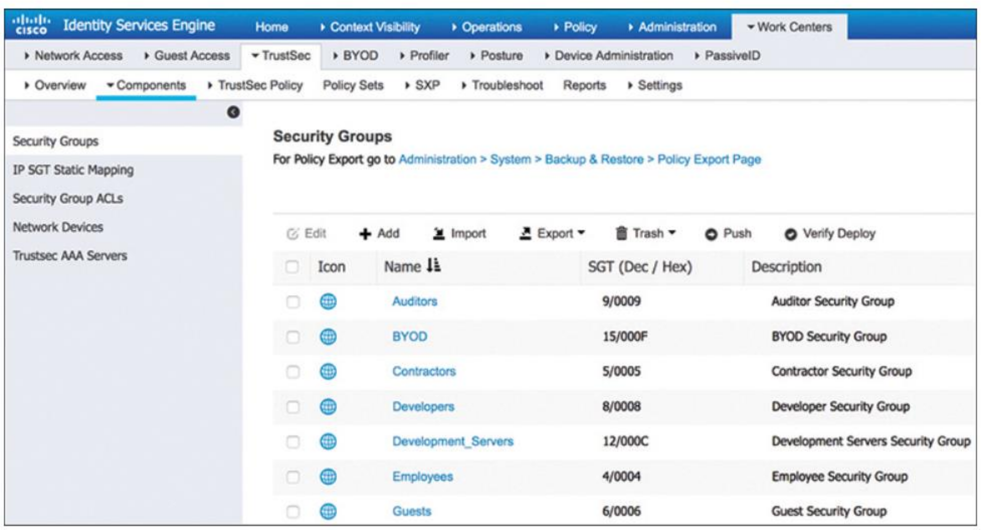
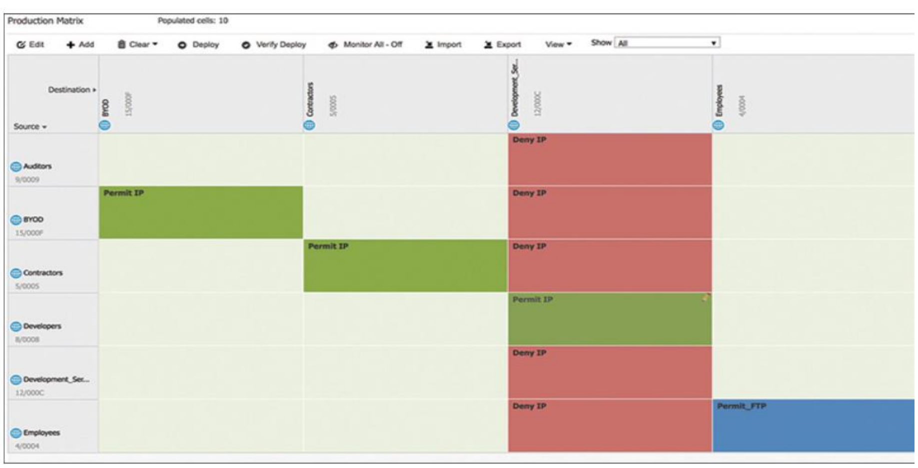
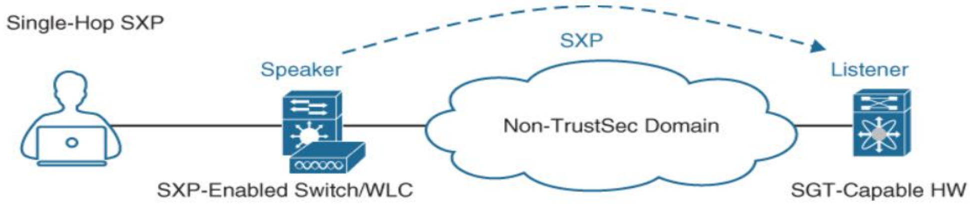
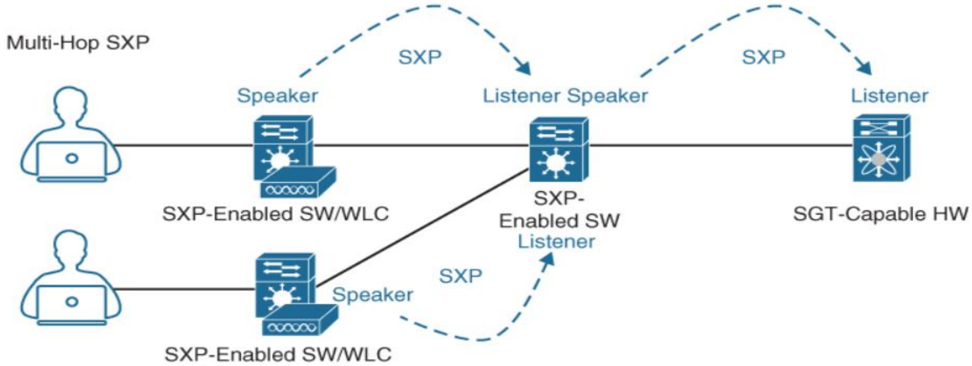
TrustSec:

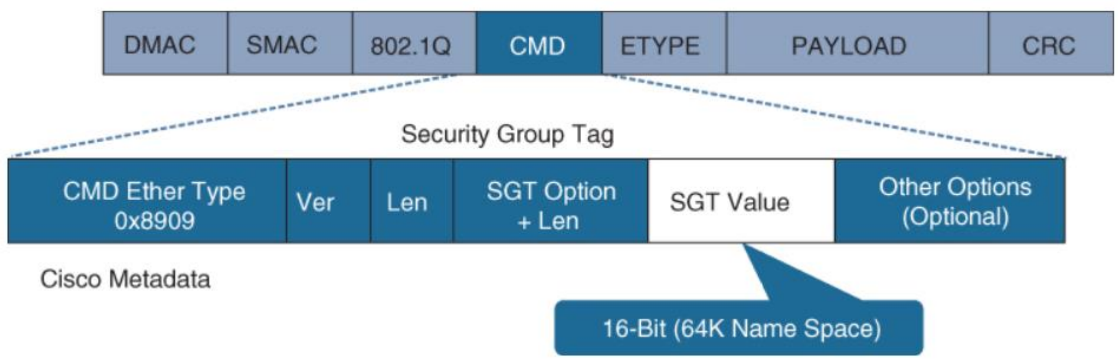
- TrustSec is next-generation access control enforcement solution developed by Cisco.
- To address the growing operational challenges related to maintaining firewall rules.
- To maintaining the Firewall rules and ACLs by using Security Group Tag (SGT) tags.
- It uses SGT tags perform ingress tagging & egress filtering enforce access control policy.
- Cisco ISE assigns the SGT tags to users or devices that are successfully authenticated.
- Cisco ISE assigns the SGT tags to users authorized through 802.1x, MAB, or WebAuth.
- The SGT tag assignment is delivered to the authenticator as an authorization option.
- After SGT tag is assigned, an access enforcement policy allow or drop based on SGT tag.
- After SGT tag is assigned SGT Tag can be applied at any egress point of TrustSec network.
- The SGT tags represent the context of the user, device, use case, or the function.
- This means SGT tags are often named after particular roles or business use cases.
- The SGT name is available on Cisco ISE and the network devices to create policies.
- What is actually inserted into a Layer 2 frame SGT tag is basically a numeric value.
- It Config occurs three phases Ingress classification, Propagation & Egress enforcement.

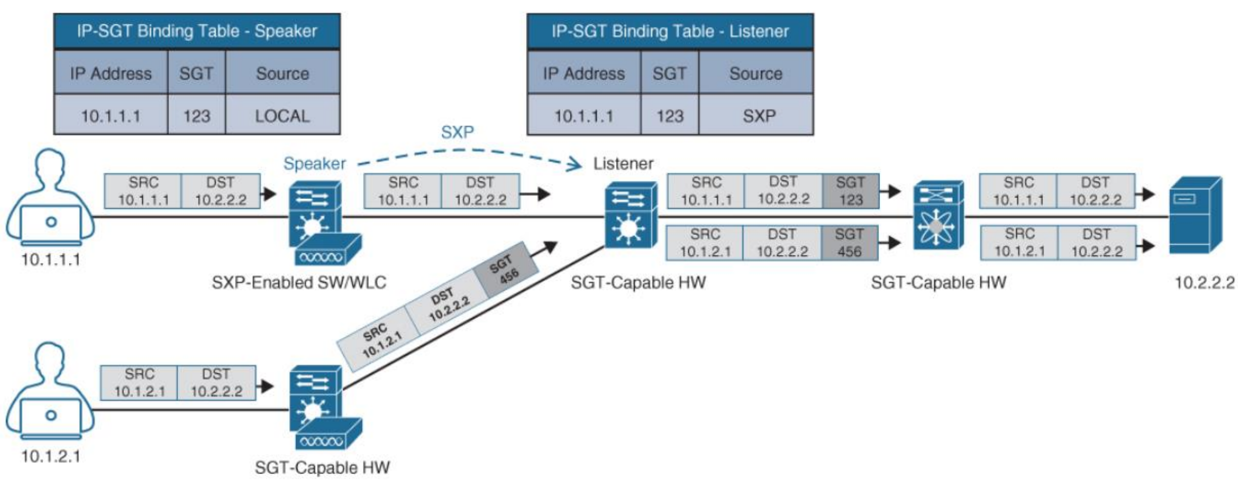





Single-Hop SXP
SXP
Speaker
Listener
Non-TrustSec Domain
SXP-Enabled Switch/WLC
SGT-Capable HW


Multi-Hop SXP
SXP
SXP
Speaker
Listener Speaker
Listener
SXP-Enabled SW/WLC
SXP-
Enabled SW
Listener
SGT-Capable HW
Speaker
SXP
SXP-Enabled SW/WLC


DMAC
SMAC
802.1Q
CMD
ETYPE
PAYLOAD
CRC
Security Group Tag
CMD Ether Type
0x8909
Ver
Len
SGT Option
+ Len
SGT Value
Other Options
(Optional)
Cisco Metadata
16-Bit (64K Name Space)


IP-SGT Binding Table - Speaker
IP Address | SGT | Source
10.1.1.1 | 123 | LOCAL
IP-SGT Binding Table - Listener
IP Address | SGT | Source
10.1.1.1 | 123 | SXP
SXP
Speaker
Listener
SRC 10.1.1.1 DST 10.2.2.2
SRC 10.1.1.1 DST 10.2.2.2
SRC 10.1.1.1 DST 10.2.2.2 SGT 123
SRC 10.1.2.1 DST 10.2.2.2 SGT 456
SRC 10.1.1.1 DST 10.2.2.2
SRC 10.1.2.1 DST 10.2.2.2
10.1.1.1
SXP-Enabled SW/WLC
SGT-Capable HW
SGT-Capable HW
10.2.2.2
SRC 10.1.2.1 DST 10.2.2.2 SGT 456
SRC 10.1.2.1 DST 10.2.2.2
10.1.2.1
SGT-Capable HW

MACsec:

- MACsec is an IEEE 802.1AE standards-based Layer 2 hop-by-hop encryption method.
- In MACsec the traffic is encrypted only on the wire between two MACsec peers.
- In MACsec the traffic is unencrypted as it is processed internally within the switch.
- This allows the Cisco switch to look into the inner packets for things like SGT tags.
- To perform packet enforcement or Quality of Service (QoS) etc prioritization.
- MACsec also leverages onboard ASICs to perform the encryption and decryption.
- Rather than having to offload to a crypto engine, as with Internet Protocol Security.
- It is based on Ethernet frame format; an additional 16-byte MACsec Security Tag field.
- MACsec provides authentication using Galois Method Authentication Code (GMAC).
- Authenticated encryption using Galois/Counter Mode Advanced Encryption Standard.